The bubble on the dumbbell as a quantics tensor network

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I. THE DUMBBELL FAMILY

The spectrum of the Green function is a segment of the real axis, and the Fermi function, or the Bose function for bosons, has its poles at the Matsubara frequencies, on the imaginary axis at distances from the origin set by the temperature. Between the segment and the poles the integrands are analytic. We take the contours from the level curves of a conformal map ω that sends a strip, periodic in the coordinate ϕ along it and with the coordinate λ across it, $\zeta = \phi + i\tau\lambda$, onto the plane cut along the segment and along the imaginary axis beyond the first poles: the line $\lambda = 0$ goes to the segment and the line $\lambda = 1$ to the imaginary axis beyond the first poles [Fig. 1(a)]. Every horizontal line in between goes to a closed curve C_λ around the segment that leaves out every pole, the contour at level λ [Fig. 1(b)]. At the imaginary axis the region is squeezed between the segment and the first poles, so the curves are narrow there and bulge on either side, like dumbbells.

On every contour ϕ is a periodic coordinate, so a function analytic in the region has Fourier coefficients in ϕ that decay exponentially, and the trapezoidal rule on a uniform grid in ϕ converges exponentially (Appendix G). Moving from one contour to another is a shift of the strip by $i\tau\Delta\lambda$, which multiplies the n-th Fourier coefficient by $e^{-\alpha n\Delta\lambda}$, with $\alpha = \pi\tau/4$ [Eq. (H1), Appendix H], so a change of level is diagonal in Fourier space. We consider a function with poles on a segment $[-E, E]$ of the real axis and the Fermi or Bose function, with poles at the Matsubara frequencies $i\omega_n = i(2n+1)\pi/\beta$ or $i\nu_m = 2\pi im/\beta$. The temperature fixes the map through the first poles, where the curve $\lambda = 1$ begins.

For a segment $[-E, E]$ and the inverse temperature β, the map is

$$\omega(\zeta) = E\kappa' \operatorname{sd}\Big(\frac{K}{2}(\zeta+1)\,\Big|\,\kappa^2\Big), \qquad \tau = \frac{2K'}{K}, \tag{1}$$

with $\kappa/\kappa' = E\beta/\pi$, $\kappa' = \sqrt{1-\kappa^2}$, $\operatorname{sd} = \operatorname{sn}/\operatorname{dn}$, and K and K' the complete elliptic integrals of the first kind with parameters κ^2 and κ'^2. We write $C_\lambda(E, \beta)$ for its level curves. As an example, Fig. 1 shows the dumbbell family for $E = 3$ and $\beta = \pi$, that is $\kappa = 3/\sqrt{10}$, at the levels $\lambda = 0.1, \ldots, 0.9$. This family avoids the poles at the fermionic Matsubara frequencies and encloses the segment $[-3, 3]$. For $\lambda \to 0$ it reduces to the segment of the real axis, and for $\lambda \to 1$ to the imaginary axis beyond the first poles $\pm i$.

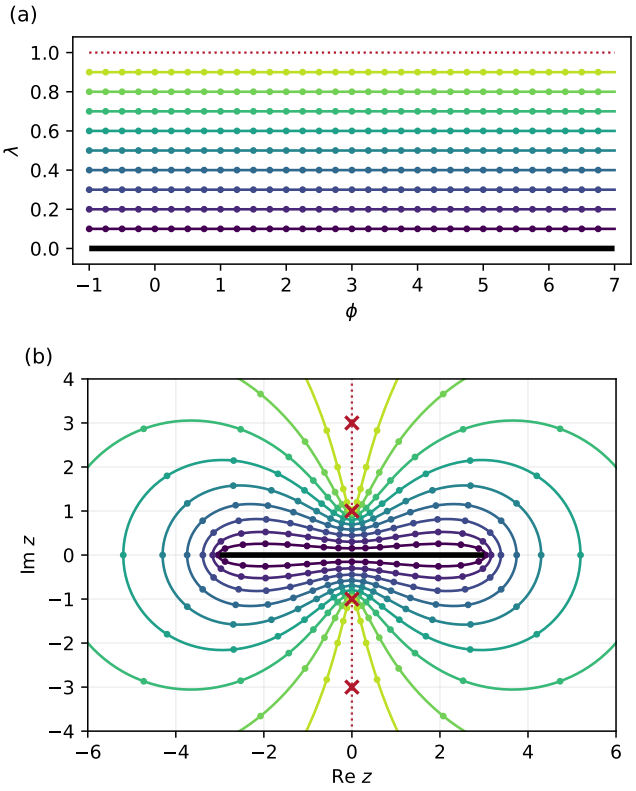


FIG. 1. The dumbbell family. (a) The strip $\zeta = \phi + i\tau\lambda$: the levels $\lambda = 0.1, \ldots, 0.9$ (colored lines) between the band ($\lambda = 0$, black) and the level $\lambda = 1$ (dotted) and the grid $\phi_j = -1 + 8j/N$ for $N = 32$ (dots). (b) Its image under ω: the dumbbells C_λ (same colors) around the band $[-3, 3]$ and the nodes $z_j = \omega_\lambda(\phi_j)$. The poles of the Fermi function at $\pm i, \pm 3i$ ($\beta = \pi$) lie on the level $\lambda = 1$, the imaginary axis beyond $\pm i$ (dotted).

A. Properties of the dumbbell map

We write $\omega_\lambda(\phi) = \omega(\phi + i\tau\lambda)$: ϕ is the position along a contour and λ its *level*, where $\tau = K'(k)/K(k)$, with $k = (L-1)/(L+1)$ and $L = \sqrt{1 + (E\beta/\pi)^2}$. The function sd satisfies $\operatorname{sd}(u + 2K) = -\operatorname{sd}(u)$, $\operatorname{sd}(u + 2iK') = -\operatorname{sd}(u)$ and $\operatorname{sd}(-u) = -\operatorname{sd}(u)$, and its Taylor coefficients are real, so

$$\begin{aligned} \omega(\zeta + 8) &= \omega(\zeta), & \omega(\zeta + 4) &= -\omega(\zeta), \\ \omega(2 - \zeta) &= \omega(\zeta), & \overline{\omega(\zeta)} &= \omega(\bar\zeta) = \omega(2 - \bar\zeta). \end{aligned} \tag{2}$$

In terms of ϕ and λ: period 8 in ϕ; $\omega_\lambda(\phi + 4) = -\omega_\lambda(\phi)$; and $\omega_\lambda(2 - \phi) = \overline{\omega_\lambda(\phi)}$, so the lower half of every contour

is the mirror image of its upper half.

For fixed $\lambda \in (0,1)$, $\phi \mapsto \omega_\lambda(\phi)$, $\phi \in [-1,7)$, traces a closed curve C_λ once, the *dumbbell at level* λ. Its shape follows from the values at $\lambda \to 0$ and $\lambda = 1$:

- $\lambda \to 0^+$: ω is real and runs $0 \to E \to 0 \to -E \to 0$ for $\phi = -1 \to 1 \to 3 \to 5 \to 7$. The contour collapses onto the segment $[-E, E]$: $\phi \in (-1,1) \cup (5,7)$ is the upper half ($\mathrm{Im}\,\omega > 0$), $\phi \in (1,5)$ the lower half. The *tips* are $\phi = 1$ and $\phi = 5$ ($\omega = \pm E$), and the curve crosses the imaginary axis at $\phi = -1 \equiv 7$ and $\phi = 3$ ($\omega = 0$). In increasing ϕ the curve is traversed clockwise.

- $\lambda = 1$: $\omega(\phi + i\tau)$ is purely imaginary with $|\mathrm{Im}\,\omega| \geq \pi/\beta$ ($\omega(-1+i\tau) = i\pi/\beta$, and $\omega \to \infty$ at $\phi = 1, 5$, where sd has its poles). The level-1 "contour" is the imaginary axis beyond the first poles, $\{\mathrm{Re}\,z = 0,\ |\mathrm{Im}\,z| \geq \pi/\beta\}$.

Every level $0 < \lambda < 1$ lies between the segment and the level-1 curve. The strip $0 < \lambda < 2$ is mapped by ω onto $\mathbb{C} \setminus [-E, E]$ (the level $2 - \lambda$ gives the mirror image $-\overline{C_\lambda}$ of C_λ, the same curve), and $\lambda = 2$ is the segment again.

The map is tied to the temperature through the poles of the Fermi function $f(z) = 1/(e^{\beta z}+1)$, at the fermionic Matsubara frequencies $i\omega_n = i(2n+1)\pi/\beta$ (Sec. II A). With $\kappa/\kappa' = E\beta/\pi$ the two closest to the real axis, $\pm i\pi/\beta$, sit at the ends of the level-1 curve, and then all poles lie on it. This is the choice of modulus of Lin, Lu, Ying and E, $L = \sqrt{1 + (E\beta/\pi)^2}$. Every contour C_λ with $0 < \lambda < 1$ separates the segment from all poles of f.

The Jacobian of the map follows from $v = \frac{K}{2}(\zeta + 1)$ and $\frac{d}{dv}\,\mathrm{sd}\,v = \mathrm{cn}\,v/\,\mathrm{dn}^2\,v$:

$$J(\zeta) \equiv \frac{\partial \omega}{\partial \phi} = E\kappa' \frac{K}{2} \frac{\mathrm{cn}\,v}{\mathrm{dn}^2\,v}, \qquad \frac{\partial \omega}{\partial \lambda} = i\tau\, J,$$
$$J^2 = \Big(\frac{K\kappa'\beta}{2\pi}\Big)^2 \Big(\frac{\pi^2}{\beta^2} + \omega^2\Big)(E^2 - \omega^2), \tag{3}$$

the second relation because ω is analytic in $\zeta = \phi + i\tau\lambda$, the third from $\mathrm{dn}^2\,v = 1/(1 + \beta^2\omega^2/\pi^2)$ and $\mathrm{cn}^2\,v = (1 - \omega^2/E^2)/(1 + \beta^2\omega^2/\pi^2)$. We write $J_\lambda(\phi) = J(\phi + i\tau\lambda)$. For $0 < \lambda < 1$, J_λ has no zeros on C_λ; on the segment it is $J_0(\varepsilon) = \frac{K\kappa'\beta}{2\pi}\sqrt{(E^2 - \varepsilon^2)(\pi^2/\beta^2 + \varepsilon^2)}$, which vanishes like $\sqrt{E - |\varepsilon|}$ at the edges, so the nodes cluster at the edges of the segment like Chebyshev points [Fig. 1(b)].

Every z off the segment and the level-1 curve lies on exactly one dumbbell $C_{\lambda'}$, and we call $\lambda(z) \equiv \lambda'$ the *level* of z.

C_λ crosses the imaginary axis at $\omega_\lambda(-1) = iE\kappa'\,\mathrm{sd}(K'\lambda\,|\,\kappa'^2)$ and $\omega_\lambda(3) = -\omega_\lambda(-1)$, by $\mathrm{sd}(iy\,|\,\kappa^2) = i\,\mathrm{sd}(y\,|\,\kappa'^2)$; these are the lowest point of its upper half and the highest point of its lower half. The highest and lowest points of C_λ follow from (3) (Fig. 2). Along C_λ the tangent J is horizontal where J^2 is real and positive. Since $J^2 \propto (\pi^2/\beta^2 + \omega^2)(E^2 - \omega^2)$, the imaginary part of J^2 for $\omega = x + iy$ is proportional to $2xy[E^2 - \pi^2/\beta^2 - 2(x^2 - y^2)]$, which off the axes vanishes on the hyperbola $x^2 - y^2 = c^2$, $c^2 = (E^2 - \pi^2/\beta^2)/2$,

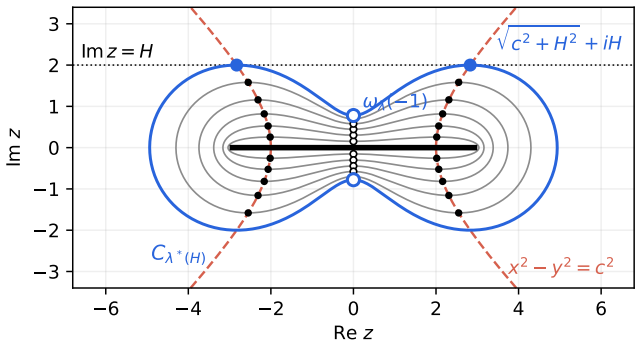


FIG. 2. Highest and lowest points of the dumbbells, for $E = 3$ and $\beta = \pi$. The extreme points (dots) of every C_λ (grey) lie on the hyperbola $x^2 - y^2 = c^2$ (dashed), where the tangent J is horizontal, and the open circles mark where they cross the imaginary axis. The dumbbell $C_{\lambda^*(H)}$ (blue) touches the line $\mathrm{Im}\,z = H$ (dotted) at its highest points $\pm\sqrt{c^2 + H^2} + iH$, here for $H = \nu_1$, and it crosses the imaginary axis at $\pm\omega_\lambda(-1) = \pm iE\kappa'\,\mathrm{sd}(K'\lambda\,|\,\kappa'^2)$ (blue open circles); every C_λ with $\lambda < \lambda^*(H)$ stays below the line.

where J^2 is positive. The highest and lowest points of a dumbbell of height H are therefore

$$z = \pm\sqrt{c^2 + H^2} \pm iH,$$

and since the dumbbells are nested, C_λ stays within $|\mathrm{Im}\,z| < H$ exactly when $\lambda < \lambda^*(H) \equiv \lambda\big(\sqrt{c^2 + H^2} + iH\big)$. We call H_λ the height of C_λ, so that its top is $\sqrt{c^2 + H_\lambda^2} + iH_\lambda$ and $\lambda^*(H_\lambda) = \lambda$.

The contour formula (16) on the nodes z_j needs G at the shifted points $z_j \pm w_k$, which lie on the translated copies $C_\lambda \pm w_k$ of the contour. Each shifted point has a level; the lowest one,

$$\lambda'_{\min}(\lambda, \mu) = \min_{j,k}\ \lambda\big(z_j \pm w_k\big), \tag{4}$$

measures how close the copies come to the segment ($\lambda'_{\min} > 0$) and whether they stay outside C_λ ($\lambda'_{\min} > \lambda$). By the symmetry $w_{k+N/2} = -w_k$ the sets $\{z_j + w_k\}$ and $\{z_j - w_k\}$ coincide. The copies come closest to the segment when w_k is where C_μ crosses the imaginary axis and z_j at the highest or lowest point of C_λ. Raising μ or lowering λ raises $\lambda'_{\min}$, and $\lambda'_{\min} > \lambda$ holds for μ above a threshold $\mu_{\mathrm{fwd}}(\lambda)$, roughly for $\lambda \lesssim \mu/2$.

II. THE FORMULA BEHIND THE NETWORK

A. The bubble

Momenta are on the grid $q = 2\pi m/N_q$, $m = 0, \ldots, N_q - 1$, and all momentum sums and shifts are modulo 2π. The temperature is $1/\beta$; the fermionic and bosonic Matsubara frequencies are

$$\omega_n = \frac{(2n+1)\pi}{\beta}, \qquad \nu_m = \frac{2\pi m}{\beta}, \qquad n, m \in \mathbb{Z}, \tag{5}$$

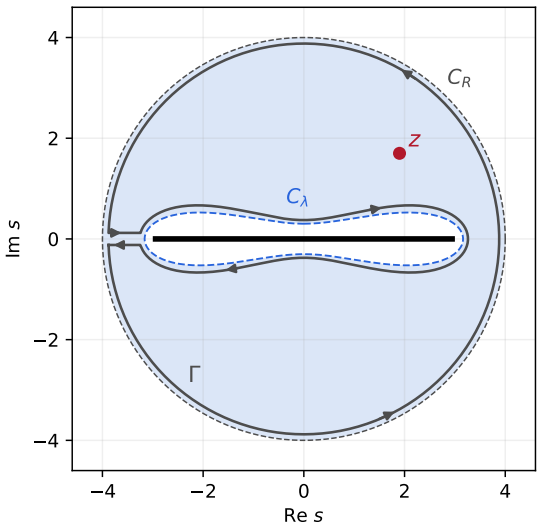


FIG. 3. Cauchy's integral formula, the first equality in (9), for G at a point z outside the dumbbell C_λ (dashed, $\lambda = 0.2$). G is analytic in the shaded region between C_λ and the circle C_R (dashed), and its singularities lie on the band $[-3,3]$ (black). A horizontal cut on the left, from C_λ to C_R, makes the region simply connected. Its counterclockwise boundary Γ (dark), drawn slightly inside it, is a single closed contour that encloses z and no singularity of G, so Cauchy's formula on it gives $G(z)$. The two sides of the cut are traversed in opposite directions and cancel, and Γ reduces to C_R counterclockwise and C_λ clockwise; the clockwise orientation turns $1/(s-z)$ into the $1/(z-s)$ of (9). As $R\to\infty$ the C_R integral vanishes.

and the Fermi function is

$$f(z) = \frac{1}{e^{\beta z}+1}, \tag{6a}$$

$$f(z+i\nu_m) = f(z), \tag{6b}$$

$$\operatorname*{Res}_{z=i\omega_n} f = -\frac{1}{\beta}. \tag{6c}$$

The one-particle Green function $G_q(z)$ is analytic off the band $[-E,E]$, and $G_q(z)\sim 1/z$ for $z\to\infty$. For any function ψ analytic inside and on C_λ, shrinking C_λ onto the band, where it runs along $\varepsilon - i0^+$ from left to right and back along $\varepsilon + i0^+$, gives

$$\frac{1}{2\pi i}\oint_{C_\lambda} ds\, G_q(s)\,\psi(s) = \int_{-E}^{E} d\varepsilon\, A_q(\varepsilon)\,\psi(\varepsilon), \tag{7}$$

with the spectral function

$$\begin{aligned} A_q(\varepsilon) &= \frac{G_q(\varepsilon - i0^+) - G_q(\varepsilon + i0^+)}{2\pi i} \\ &= -\frac{1}{\pi}\operatorname{Im} G_q(\varepsilon + i0^+), \end{aligned} \tag{8}$$

the jump of G_q across the band. For z outside C_λ, $\psi(s) = 1/(z-s)$ gives the spectral representation

$$G_q(z) = \frac{1}{2\pi i}\oint_{C_\lambda} ds\,\frac{G_q(s)}{z-s} = \int_{-E}^{E} d\varepsilon\,\frac{A_q(\varepsilon)}{z-\varepsilon}, \tag{9}$$

where the first equality is Cauchy's formula on the contour Γ of Fig. 3, which consists of a circle C_R, counterclockwise, and of C_λ, clockwise; the C_R integral vanishes as $R\to\infty$. With $\psi(s) = e^{-ist}$, (7) gives the time representation, the real-time Green function

$$\bar g_q(t) = \frac{1}{2\pi i}\oint_{C_\lambda} ds\, G_q(s)\, e^{-ist} = \int_{-E}^{E} d\varepsilon\, A_q(\varepsilon)\, e^{-i\varepsilon t}, \tag{10}$$

and $\psi(s) = f(s)\, e^{ist}$ gives the occupied real-time Green function $g_q(t)$, obtained from $\bar g_q(-t)$ by replacing G_q with fG_q and A_q with fA_q.

As an example we take a tight-binding chain with nearest-neighbour hopping,

$$\varepsilon_q = 3\cos q, \qquad G_q(z) = \frac{1}{z-\varepsilon_q}, \qquad A_q(\varepsilon) = \delta(\varepsilon - \varepsilon_q), \tag{11}$$

at the inverse temperature $\beta = \pi$; its band is $[-3,3]$, and the bubble has its cut on $[-6,6]$. For this model the bubble is the Lindhard function (F3), the reference for every accuracy quoted.

The bubble reads

$$\begin{aligned} \chi_Q(i\nu_m) &= \frac{1}{N_q\beta}\sum_{q,n} G_q(i\omega_n)\, G_{q+Q}(i\omega_n + i\nu_m) \\ &= \frac{1}{N_q}\sum_q \frac{1}{\beta}\sum_n h(i\omega_n), \end{aligned} \tag{12}$$

with

$$h(z) = G_q(z)\, G_{q+Q}(z + i\nu_m). \tag{13}$$

The Matsubara sum converges absolutely because the summand is $O(\omega_n^{-2})$. By (9) the poles of h, which merge into cuts for a continuous spectrum, lie on the band $[-E,E]$ of the real axis, from $G_q(z)$, and on the band moved down by ν_m, from $G_{q+Q}(z+i\nu_m)$. These lie at the heights 0 and $-\nu_m$, even multiples of π/β, while the poles $i\omega_n$ of f lie at odd multiples. So h is regular at the poles of f, and with the residue $-1/\beta$ of f in (6c), $\operatorname{Res}_{z=i\omega_n}[f(z)h(z)] = -h(i\omega_n)/\beta$. Hence

$$\frac{1}{\beta}\sum_n h(i\omega_n) = -\sum_n \operatorname*{Res}_{z=i\omega_n}\big[f(z)\,h(z)\big]. \tag{14}$$

B. The contour formula

Let C_R be the counterclockwise circle centred at the origin with radius R, halfway between two poles of f. For $R\to\infty$ it contains both the poles of h, at $\operatorname{Im} z = 0$ and $\operatorname{Im} z = -\nu_m$, and those of f, and elsewhere inside it fh is analytic. For $m>0$, as shown in Fig. 4, the contour integral over C_R can be rewritten as a sum of contour integrals over $C_R = C_- + C_1 + C_2 + C_3 + C_+$. These are the counterclockwise boundaries of the five regions into which

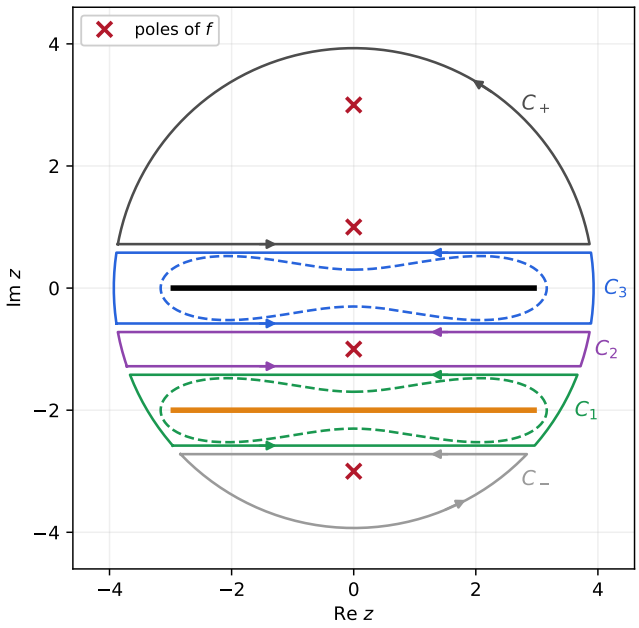


FIG. 4. The decomposition of the circle C_R that leads to (15), for $\beta = \pi$ and $m = 1$ ($\nu_1 = 2$). The poles of h of (13) lie on the band $[-3, 3]$ of the real axis (black), from $G_q(z)$, and on the band moved down by ν_m (orange), from $G_{q+Q}(z + i\nu_m)$. Horizontal lines at the distance η above and below them cut the disk into five regions, whose counterclockwise boundaries C_+, C_3, C_2, C_1 and C_- are drawn slightly inside them. C_3 encloses the band, C_1 the shifted band, and C_+, C_2, C_- the poles of f. Each line is traversed in both directions, so the five contours add up to C_R. The dashed curves are the dumbbells C_λ and $C_\lambda - i\nu_m$, here for $\lambda = 0.2$, into which C_3 and C_1 are deformed.

horizontal lines, at the distance $\eta < \pi/\beta$ above and below the poles of h, cut the disk; each line is traversed twice, in opposite directions, and drops out. C_3 encloses the poles of h on the real axis and C_1 those on the shifted line, while C_-, C_2 and C_+ enclose only poles of f. As $R \to \infty$, the integrals of $fh/(2\pi i)$ over C_-, C_2 and C_+ give minus the right-hand side of (14), while the integral over C_R vanishes, because $h = O(z^{-2})$ and f stays bounded on C_R, so that

$$\sum_n \operatorname*{Res}_{z=i\omega_n} \big[f(z)\,h(z)\big] = -\frac{1}{2\pi i}\oint_{C_1} f\,h\,dz - \frac{1}{2\pi i}\oint_{C_3} f\,h\,dz. \tag{15}$$

The contour integrals over C_3 and C_1 can be deformed into integrals over the dumbbells C_λ and $C_\lambda - i\nu_m$ (dashed lines in Fig. 4) as long as C_λ leaves out the shifted band $[-E, E] - i\nu_m$. The dumbbell C_λ stays within $|\mathrm{Im}\, z| < \nu_m$ for $\lambda < \lambda^*(\nu_m)$ (Sec. I A), so the deformations are safe for $\lambda < \lambda^*(\nu_m)$, and since $\nu_m \geq \nu_1$, the bound for $m = 1$ covers every $m > 0$. By the change of variables $z \to z - i\nu_m$ in the integral over $C_\lambda - i\nu_m$

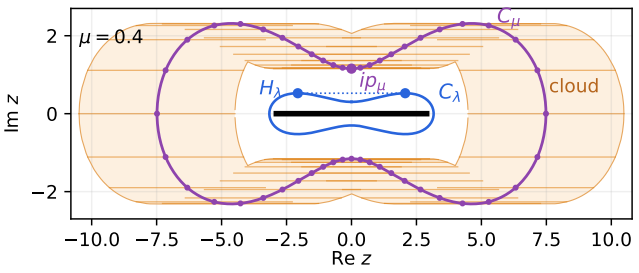


FIG. 5. The condition $\lambda < \Lambda(\mu, \beta)$ for $\lambda = 0.2$, $\mu = 0.4$ and $\beta = \pi$, with the vertical scale stretched 1.5 times. The thin lines are the shifted cuts $[-3, 3] \pm w$ for points w on C_μ (dots); for all w on C_μ they sweep the cloud $[-3, 3] \pm C_\mu$ (shaded), the zone that C_λ must avoid. Near the band its edge comes from the points $\pm ip_\mu$ where C_μ crosses the imaginary axis, and the dot on C_λ is its highest point. The point ip_μ lies above the highest point of C_λ, so C_λ lies in the hole of the cloud and (16) gives $\chi_Q(w_k)$.

and by (6b), we have, at $w = i\nu_m$,

$$\chi_Q(w) = \frac{1}{N_q}\sum_q \frac{1}{2\pi i}\oint_{C_\lambda} dz\; f(z)\,G_q(z) \times \Big[G_{q+Q}(z + w) + G_{q-Q}(z - w)\Big]. \tag{16}$$

At $w = i\nu_m$ and for $\lambda < \lambda^*(\nu_m)$, the factors $f(z)\,G_{q\pm Q}(z \pm w)$ are analytic inside and on C_λ. Inserting (9), Cauchy's formula turns (16) into

$$\chi_Q(w) = \frac{1}{N_q}\sum_q \int_{-E}^{E} d\varepsilon \int_{-E}^{E} d\varepsilon'\, A_q(\varepsilon)\, A_{q+Q}(\varepsilon') \times \frac{f(\varepsilon) - f(\varepsilon')}{w + \varepsilon - \varepsilon'}, \tag{17}$$

with $w = i\nu_m$. Equation (17) does not depend on λ, is analytic in w off the interval $[-2E, 2E]$ of the differences $\varepsilon' - \varepsilon$ and vanishes as $|w| \to \infty$, so it is the analytic continuation $\chi_Q(w)$ of the bubble.

Now let w be a general complex number. The same steps give (17) as long as C_λ leaves out the shifted cuts $[-E, E] \mp w$, where $G_{q\pm Q}(z \pm w)$ is singular. For $w = w_k \in C_\mu$, where $C_\mu = C_\mu(2E, \beta/2) = 2\,C_\mu(E, \beta)$ is a dumbbell which encloses $[-2E, 2E]$ and leaves out the poles $i\nu_m$, $m \neq 0$, of the Bose function, the shifted cuts sweep the set $[-E, E] \pm C_\mu$, the "cloud" in Fig. 5, so (16) holds on C_μ when the cloud lies outside C_λ. This happens when the top of C_λ lies below the point ip_μ where C_μ crosses the imaginary axis, with $p_\mu = 2\,\mathrm{Im}\,\omega(-1 + i\tau\mu)$ (Sec. I A):

$$H_\lambda < p_\mu = 2E\kappa'\,\mathrm{sd}(K'\mu\,|\,\kappa'^2).$$

In terms of levels this reads $\lambda < \Lambda(\mu, \beta)$, where $\Lambda(\mu, \beta) = \lambda^*(p_\mu) = \lambda\big(\sqrt{c^2 + p_\mu^2} + ip_\mu\big)$. Since $p_\mu < \nu_1 \leq \nu_m$ and the level $\lambda^*(H)$ grows with H, $\Lambda(\mu, \beta) = \lambda^*(p_\mu) < \lambda^*(\nu_m)$, and the only condition on λ becomes $\lambda < \Lambda(\mu, \beta)$.

C. The bubble as a product in time

The data are the values of G on C_λ, while (16) needs $G_{q\pm Q}$ at the shifted points $z \pm w$, off the contour. A time representation of (17) avoids them. Since ε and ε' are real,

$$\frac{1}{w+\varepsilon-\varepsilon'} = \begin{cases} -i\displaystyle\int_0^\infty dt\, e^{i(w+\varepsilon-\varepsilon')t}, & \operatorname{Im} w > 0, \\ i\displaystyle\int_0^\infty dt\, e^{-i(w+\varepsilon-\varepsilon')t}, & \operatorname{Im} w < 0. \end{cases} \tag{18}$$

Inserted into (17) with $\operatorname{Im} w > 0$, the first line makes the integrals over ε and ε' factorize into the real-time Green functions $\bar g_q$ of (10) and g_q. The term with $f(\varepsilon)$ gives $g_q(t)\,\bar g_{q+Q}(t)$, and the term with $f(\varepsilon')$ gives $\bar g_q(t)^*\, g_{q+Q}(t)^*$, so the bubble becomes a product in time,

$$\begin{aligned} \chi_Q(w) = &-\frac{i}{N_q}\sum_q \int_0^\infty dt\, e^{iwt} \\ &\times \big[g_q(t)\,\bar g_{q+Q}(t) - \bar g_q(t)^*\, g_{q+Q}(t)^* \big], \end{aligned} \tag{19}$$

Since $\bar g_q$ and g_q are integrals of the data over C_λ, G is never needed off the contour. On the lower half of C_μ the second line of (18) gives the same with all functions conjugated, $\chi_Q(w) = \chi_Q(\bar w)^*$. Near the tips of C_μ, where $\operatorname{Im} w \to 0$, the time integral needs long times, and the contour sums for g_q and $\bar g_q$, whose terms grow like $e^{H_\lambda t}$, lose accuracy; there the Cauchy route of Appendix A applies.

D. The time domain on the nodes

The real-time functions g_q and $\bar g_q$ are integrals over C_λ. Parametrize C_λ by ϕ: $z = \omega_\lambda(\phi)$, $dz = J_\lambda(\phi)\, d\phi$. The parametrization is clockwise, so for any F analytic near the contour

$$\frac{1}{2\pi i}\oint_{C_\lambda} F(z)\, dz = -\frac{1}{2\pi i}\int_{-1}^{7} d\phi\; J_\lambda(\phi)\, F\big(\omega_\lambda(\phi)\big). \tag{20}$$

The integrand is 8-periodic in ϕ, and the trapezoidal rule on the uniform grid $\phi_j = -1 + j\,\Delta\phi$, $j = 0, \dots, N-1$, with $\Delta\phi = 8/N$, is $\Delta\phi \sum_j$. The nodes are $z_j = \omega_\lambda(\phi_j)$, and the bubble is computed at the points $w_k = 2\,\omega(\phi_k + i\tau\mu)$ of C_μ on the same grid. With the Fermi function in the integrand it gives

$$\begin{aligned} &\frac{1}{2\pi i}\oint_{C_\lambda} f\, F\, dz \approx \sum_{j=0}^{N-1} c_j\, F(z_j), \\ &c_j = d_j\, f(z_j), \qquad d_j = -\frac{\Delta\phi}{2\pi i} J_\lambda(\phi_j). \end{aligned} \tag{21}$$

For every F that is analytic outside C_λ and $O(z^{-2})$ at infinity, the residue theorem between C_λ and a large circle, as in Sec. II B, gives $\sum_j c_j F(z_j) \approx \beta^{-1} \sum_n F(i\omega_n)$: the N nodes with the weights c_j replace the Matsubara frequencies. Without the Fermi function the rule reads $\frac{1}{2\pi i}\oint_{C_\lambda} F\, dz \approx \sum_m d_m\, F(z_m)$. The c_j depend on j, λ and β only, never on w, q or G, and the Fermi function is evaluated only at the nodes of C_λ. Both sets of weights are smooth in ϕ_j (rank 6 to 8 as quantics tensor trains, Appendix B). The quadrature error falls exponentially with N, like $e^{-\alpha N d}$ (G2), where d is the distance in level from C_λ to the nearest singularity of the integrand (Appendix G).

The two rules applied to g_q and $\bar g_q$ give the real-time functions from the data,

$$\begin{aligned} g_q(t) &\approx \sum_j c_j\, G_q(z_j)\, e^{iz_j t}, \\ \bar g_q(t) &\approx \sum_m d_m\, G_q(z_m)\, e^{-iz_m t}, \end{aligned} \tag{22}$$

and the time integral in (19) is replaced by a rule $\int_0^\infty dt\, h(t) \approx \sum_p \varpi_p\, h(t_p)$ with N_t points, for instance the geometric grid $t_p = t_{\min}\rho^p$, $p = 0, \dots, N_t - 1$, with $\varpi_p = t_p \ln\rho$ (half at the two ends), the trapezoidal rule in $\ln t$. Then

$$\begin{aligned} \chi_Q(w_k) &\approx -i \sum_p \varpi_p\, e^{iw_k t_p} \big[Y_p(Q) - Y_p(-Q)^* \big], \\ Y_p(Q) &= \frac{1}{N_q}\sum_q g_q(t_p)\, \bar g_{q+Q}(t_p), \end{aligned} \tag{23}$$

since the second product in (19) is $Y_p(-Q)^*$.

A function given on the N nodes of C_λ has the discrete Fourier representation

$$\begin{aligned} G_q(z_j) &= \sum_n \hat g_{q,n}\, e^{i\pi n\phi_j/4}, \\ \hat g_{q,n} &= \frac{1}{N}\sum_{j=0}^{N-1} G_q(z_j)\, e^{-i\pi n\phi_j/4}, \end{aligned} \tag{24}$$

where n runs over any N consecutive integers (we use the two's-complement range $-N/2 \le n < N/2$); both relations are exact on the grid. Since $e^{i\pi n\phi_j/4} = e^{-i\pi n/4} e^{2\pi i n j/N}$, the coefficient $\hat g_{q,n}$ is $e^{i\pi n/4}/N$ times the standard FFT of the data.

In these modes the real-time functions are linear in the data,

$$\begin{aligned} g_q(t_p) &= \sum_n \hat g_{q,n}\, V^{(1)}_{n,p}, \\ \bar g_q(t_p) &= \sum_n \hat g_{q,n}\, V^{(2)}_{n,p}, \\ V^{(1)}_{n,p} &= \sum_j c_j\, e^{i\pi n\phi_j/4}\, e^{iz_j t_p}, \\ V^{(2)}_{n,p} &= \sum_m d_m\, e^{i\pi n\phi_m/4}\, e^{-iz_m t_p}, \end{aligned} \tag{25}$$

and only a window of modes n matters, because the coefficients decay exponentially (Appendix H). The matrices $V^{(1)}$ and $V^{(2)}$ depend only on the geometry and the

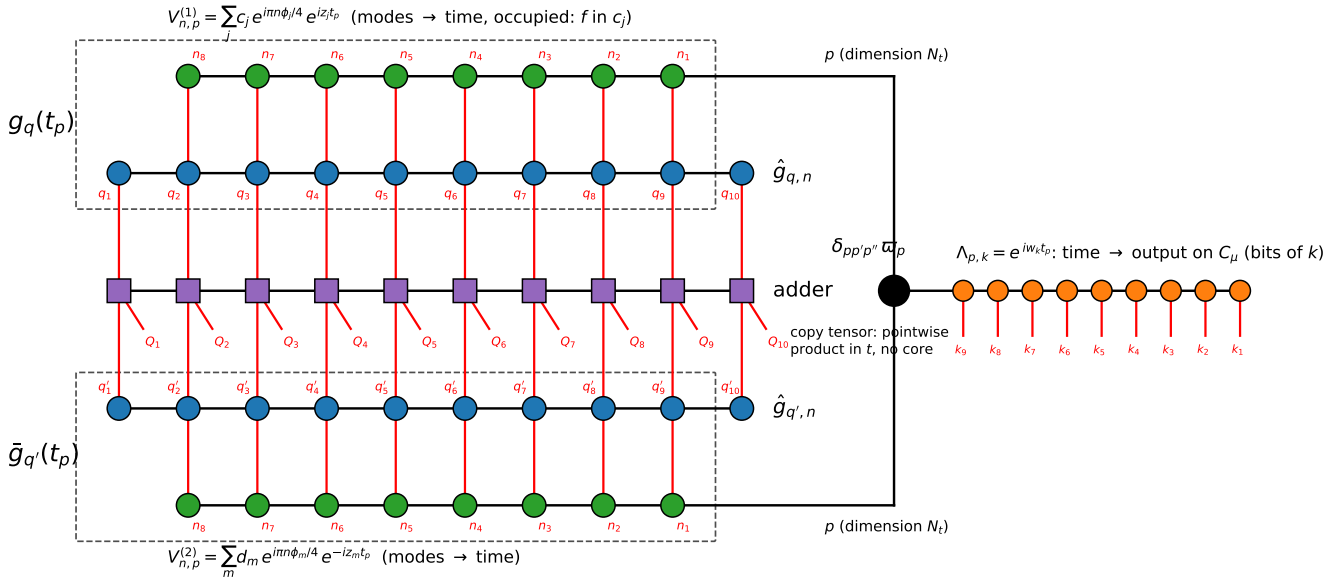


FIG. 6. The time-domain network, in the conventions of Fig. 8 (red legs are contracted, bits most significant first). Upper branch: the input train $\hat{g}_{q,n}$ (blue), its n legs contracted with the train $V^{(1)}_{n,p}$ (green), which carries the modes to the time points t_p and contains the Fermi function through c_j; the result is $g_q(t_p)$, a train over the bits of q with the time leg p. Lower branch: the same with $V^{(2)}$ (weights d_m, no f, opposite sign in the exponent), which gives $\bar{g}_{q'}(t_p)$. The adder (purple) sums q and q' with the transfer momentum Q open. The two time legs meet at a copy tensor carrying the weights ϖ_p, the product in time, and the Laplace train $\Lambda_{p,k} = e^{iw_k t_p}$ (orange) sends the time leg to the output points of C_μ (bits of k). The term $Y_p(-Q)^*$ of (23) is the same network read at $-Q$ and conjugated.

time grid. In the network of Fig. 6 the input train $\hat{g}$ is contracted with $V^{(1)}$ and $V^{(2)}$, the adder forms $Y_p(Q)$, a copy tensor multiplies by ϖ_p, and the Laplace train $\Lambda_{p,k} = e^{iw_k t_p}$ sends the time leg to the output points w_k.

Appendix A: The Cauchy route

In this appendix $G_{q\pm Q}$ at the shifted points $z \pm w$ is obtained from its values on C_λ with Cauchy's formula, instead of the time representation of Sec. II C. This route covers all of C_μ, its tips included, when the shifted points lie outside C_λ, that is $\lambda'_{\min} > \lambda$ in (4). Appendices B to E build its tensor network and test it.

Formula (16) needs $G_{q\pm Q}$ at the shifted points $z \pm w$, off the contour, while the data are the values of G on the dumbbell C_λ, and the method evaluates G only there. Cauchy's formula, the first equality in (9), gives $G(x)$ at every x outside C_λ. For x *inside* C_λ the same integral gives 0 instead of $G(x)$: then $s \mapsto G(s)/(x-s)$ is analytic outside C_λ and $O(s^{-2})$, so its integral over C_λ equals its integral over C_R, which vanishes. With $x = z \pm w$ in (16), the bubble becomes a double integral over the dumbbell, in which G enters only on C_λ:

$$\chi_Q(w) = \frac{1}{N_q} \sum_q \frac{1}{(2\pi i)^2} \oint_{C_\lambda} dz \oint_{C_\lambda} ds\, f(z)\, G_q(z) \times \left[\frac{G_{q+Q}(s)}{z - s + w} + \frac{G_{q-Q}(s)}{z - s - w} \right]. \tag{A1}$$

1. The exterior-Cauchy form

The rule $\frac{1}{2\pi i} \oint_{C_\lambda} F\, dz \approx \sum_m d_m\, F(z_m)$ of Sec. II D turns the Cauchy formula in (9) into a sum over the nodes,

$$G(x) = \sum_{m=0}^{N-1} d_m \frac{G(z_m)}{x - z_m} \qquad (x \text{ outside } C_\lambda), \tag{A2}$$

up to the trapezoidal error. The two rules applied to (A1), c_j for the integral over z and d_m for the integral over s, give the *exterior-Cauchy form* of the bubble,

$$\chi_Q(w_k) = \frac{1}{N_q} \sum_q \sum_{j,m} c_j\, G_q(z_j)\, d_m \times \left[\frac{G_{q+Q}(z_m)}{z_j - z_m + w_k} + \frac{G_{q-Q}(z_m)}{z_j - z_m - w_k} \right]. \tag{A3}$$

2. Fourier modes and the kernel

The output is expanded on C_μ in the same way as G in (24),

$$\chi_Q(w_k) = \sum_\ell \hat{\chi}_{Q,\ell}\, e^{i\pi\ell\phi_k/4}, \qquad \hat{\chi}_{Q,\ell} = \frac{1}{N} \sum_{k=0}^{N-1} \chi_Q(w_k)\, e^{-i\pi\ell\phi_k/4}. \tag{A4}$$

Insert (24) for $G_q(z_j)$ (index n') and for $G_{q\pm Q}(z_m)$ (index n) into (A3). The momentum average factorizes from the geometry:

$$\begin{aligned}\chi_Q(w_k) &= \sum_{n',n}\Big[P_Q^{+}(n',n)\,K_{n'n}(w_k)\\ &\qquad + P_Q^{-}(n',n)\,K_{n'n}(-w_k)\Big],\\ K_{n'n}(w) &= \sum_{j,m} c_j\,d_m\,\frac{e^{i\pi(n'\phi_j+n\phi_m)/4}}{z_j-z_m+w},\\ P_Q^{\pm}(n',n) &= \frac{1}{N_q}\sum_q \hat g_{q,n'}\hat g_{q\pm Q,n},\end{aligned} \tag{A5}$$

with the kernel $K_{n'n}$ and the momentum correlations $P_Q^{\pm}$. Since $\phi_{k+N/2}=\phi_k+4$ and $\omega(\zeta+4)=-\omega(\zeta)$ (2), $-w_k=w_{k+N/2}$, so the second term is the first kernel read at $k+N/2$. Project on the output modes with (A4); for the second term substitute $k'=k+N/2$ and use $\phi_k=\phi_{k'}-4$, which produces $e^{i\pi\ell}=(-1)^\ell$:

$$\begin{aligned}\hat\chi_{Q,\ell} &= \sum_{n',n}\widehat K_{n'n\ell}\big[P_Q^{+}(n',n)+(-1)^\ell P_Q^{-}(n',n)\big],\\ P_Q^{\pm}(n',n) &= \frac{1}{N_q}\sum_q \hat g_{q,n'}\,\hat g_{q\pm Q,n},\end{aligned} \tag{A6}$$

$$\begin{aligned}\widehat K_{n'n\ell} &= \frac{1}{N}\sum_k e^{-i\pi\ell\phi_k/4}\\ &\qquad\times\sum_{j,m} c_j\,d_m\,\frac{e^{i\pi(n'\phi_j+n\phi_m)/4}}{z_j-z_m+w_k}.\end{aligned} \tag{A7}$$

This is the exterior-Cauchy form (A3), where the kernel depends only on the two dumbbells and on β (through c_j); all of G, model and momentum, is in $P_Q^{\pm}$. For each k the inner double sum is a two-dimensional FFT over (j,m) of the array $c_jd_m/(z_j-z_m+w_k)$, and the outer sum is an FFT over k. In exact arithmetic the sums in (A6) run over all N values of n', n and ℓ. In practice only a window of each index matters, because the coefficients of G and of χ decay exponentially (Appendix H).

3. Compressing the kernel

The kernel (A7) is compressed into Tucker form by a truncated higher-order SVD (HOSVD). For a tensor $T_{n'n\ell}$ let $T_{(i)}$ be its mode-i unfolding (the matrix with the i-th index as rows and the other two, combined, as columns). The HOSVD takes for $U^{(i)}$ the leading left singular vectors of $T_{(i)}$ and forms the core $C = T\times_1 U^{(1)\dagger}\times_2 U^{(2)\dagger}\times_3 U^{(3)\dagger}$, i.e.

$$\begin{aligned}C_{abc} &= \sum_{n'n\ell} T_{n'n\ell}\,\overline{U^{(1)}_{n'a}}\,\overline{U^{(2)}_{nb}}\,\overline{U^{(3)}_{\ell c}},\\ T_{n'n\ell} &\approx \sum_{a=1}^{r_1}\sum_{b=1}^{r_2}\sum_{c=1}^{r_3} C_{abc}\,U^{(1)}_{n'a}\,U^{(2)}_{nb}\,U^{(3)}_{\ell c}.\end{aligned} \tag{A8}$$

Each rank r_i is the smallest with $\sum_{s>r_i}(\sigma_s^{(i)})^2 \le \frac{\varepsilon^2}{3}\|T\|_F^2$, $\sigma_s^{(i)}$ the singular values of $T_{(i)}$. Since the three truncations are orthogonal projections, their errors add at most in quadrature, $\|T-\hat T\|_F^2 \le \sum_i\sum_{s>r_i}(\sigma_s^{(i)})^2 \le \varepsilon^2\|T\|_F^2$, so ε bounds the relative error of the compressed tensor. The result is quasi-optimal: its error is at most $\sqrt{3}$ times that of the best Tucker approximation with the same ranks. Applied to the weighted kernel $\widetilde K_{n'n\ell} = s_{n'}s_n\widehat K_{n'n\ell}$ of Appendix H, whose weights s_n make a relative truncation error of $\widetilde K$ an error of the same size in $\hat\chi$,

$$s_{n'}\,s_n\,\widehat K_{n'n\ell} \approx \sum_{a,b,c} C_{abc}\,U^{(1)}_{n'a}\,U^{(2)}_{nb}\,U^{(3)}_{\ell c}, \tag{A9}$$

at $\varepsilon = 10^{-8}$ the ranks are $(r_1,r_2,r_3)=(31,33,44)$ for $(\lambda,\mu)=(0.2,0.4)$ and $(56,60,87)$ for $(0.1,0.2)$. The core has $r_1r_2r_3 = 45\,012$ entries in the example.

Inserting (A9) into (A6) and absorbing the weights into the leaves, $\tilde U^{(i)}_{na} = U^{(i)}_{na}/s_n$, gives the bubble in factorized form, the formula the network computes:

$$\begin{aligned}A_{q,a} &= \sum_n \hat g_{q,n}\,\tilde U^{(1)}_{na}, \qquad B_{q,b} = \sum_n \hat g_{q,n}\,\tilde U^{(2)}_{nb},\\ Y_c^{\pm}(Q) &= \frac{1}{N_q}\sum_q\sum_{a,b} A_{q,a}\,C_{abc}\,B_{q\pm Q,b},\\ \hat\chi_{Q,\ell} &= \sum_c U^{(3)}_{\ell c}\Big[Y_c^{+}(Q)+(-1)^\ell\,Y_c^{-}(Q)\Big],\\ \chi_Q(w_k) &= \sum_\ell \hat\chi_{Q,\ell}\,e^{i\pi\ell\phi_k/4}.\end{aligned} \tag{A10}$$

Per momentum the input is reduced to r_1+r_2 numbers; $Y^{\pm}$ are r_1r_2 cyclic correlations in q contracted with the core; the output needs r_3 modes. Nothing in (A10) depends on N except the final synthesis. For any G, the definition of $Y^{\pm}$ gives the exact identity

$$Y_c^{-}(Q) = Y_c^{+}(-Q \bmod N_q), \tag{A11}$$

since $B_{q-Q,b} = B_{q+(-Q),b}$. The branch Y^- therefore follows from Y^+ by the reflection $Q\to -Q$, without a second contraction (Appendix D), and since the reflection changes a bond by at most a factor 2, the bonds of the two branches agree within that factor. For the even band $\varepsilon_q = 3\cos q$ the two branches are equal: $A_{-q,a}=A_{q,a}$ and $B_{-q,b}=B_{q,b}$, and the substitution $q\to -q$, which maps the momentum grid onto itself, gives

$$\frac{1}{N_q}\sum_q A_{q,a}B_{q-Q,b} = \frac{1}{N_q}\sum_q A_{q,a}B_{q+Q,b}, \tag{A12}$$

that is, $Y^-(Q)=Y^+(Q)$ exactly on the grid. Then $\hat\chi_{Q,\ell} = [1+(-1)^\ell]\sum_c U^{(3)}_{\ell c}Y_c^{+}(Q)$: the odd modes vanish, which is the evenness $\chi_Q(-w)=\chi_Q(w)$ of the bubble of an even band. Computed by two separate zip-ups, each recompressed at 10^{-9}, the two trains differ by

$\|Y^- - Y^+\|/\|Y^+\| = 1.6 \times 10^{-11}$. The network does not use this symmetry: it computes both branches, as it must for a band that is not even.

The ranks are small because G and χ enter only through Cauchy transforms. Equation (17) writes the bubble as the average of $[f(\varepsilon) - f(\varepsilon')]/(w + \varepsilon - \varepsilon')$ with the weights $A_q(\varepsilon)A_{q+Q}(\varepsilon')$: a trilinear form in fA_q, A_{q+Q} and the output kernel $1/(w + \varepsilon - \varepsilon')$, and (A3) is its discretization. The first G enters only through $\sum_j c_j G_q(z_j)/(z_j - x) = \int f(\varepsilon)A_q(\varepsilon)\, d\varepsilon/(\varepsilon - x)$, the Cauchy transform of fA_q at the points $x = z_m - w_k$; the second through the Cauchy transform of $A_{q\pm Q}$ at $z_j \pm w_k$; and the output is the Cauchy transform of a two-particle spectral function on $[-2E, 2E]$, evaluated on C_μ. So each leg of $\widetilde{K}$ is a Cauchy kernel $1/(x - \varepsilon)$ between a segment and a set at a positive distance from it, with $\Lambda(\mu, \beta)$ the lowest level of the cloud $[-E, E] \pm C_\mu$ (Sec. II B):

leg	source segment	target set	lowest level of the target
1 (n')	$[-E, E]$, weight fA	$[-E, E] - C_\mu$ (the "cloud")	$\Lambda(\mu, \beta)$
2 (n)	$[-E, E]$, weight A	$[-E, E] + C_\mu$ (the same set)	$\Lambda(\mu, \beta)$
3 (ℓ)	$[-2E, 2E]$ (doubled family, level 0)	C_μ	μ

In dumbbell coordinates a Cauchy kernel from the band to a contour at level Λ is the diagonal multiplier $e^{-\alpha\Lambda n}$ of (H1), so its singular values decay exponentially at the rate $\alpha\Lambda$ and its ε-rank is $\log(1/\varepsilon)/(\alpha\Lambda)$. The discrete Lehmann representation works the same way, with the dumbbell level in the role of $\log \Lambda_{\mathrm{DLR}}$. Measured for μ from 0.2 to 0.8, the ranks follow, within 3 %,

$$r_3 \approx 0.90\, \frac{\log(1/\varepsilon)}{\alpha\mu}, \qquad r_{1,2} \approx 0.65\, \frac{\log(1/\varepsilon)}{\alpha\Lambda(\mu, \beta)}; \tag{A13}$$

for $\varepsilon = 10^{-8}$ this gives 42 and 31 at $\mu = 0.4$ (measured 44 and 31), and 84 and 53 at $\mu = 0.2$ (measured 87 and 56). The prefactors are below 1 because the HOSVD measures the error in the Frobenius norm over the whole cloud, most of which lies far above level Λ; the sup-norm version (the number of poles needed to represent a Green function on C_Λ) is $\log(1/\varepsilon)/(\alpha\Lambda)$ within 1 to 3 %. The ranks therefore grow like $\log(1/\varepsilon)$ (about +9 per two decades for $(0.2, 0.4)$ and +16 for $(0.1, 0.2)$), like $1/\mu$, and roughly linearly with β at fixed level, because at fixed level the whole dumbbell family moves toward the real axis as $T \to 0$. They do not depend on N or on λ.

Appendix B: Quantics tensor trains

An index $q \in \{0, \dots, 2^R - 1\}$ is written in binary,

$$q = \sum_{i=1}^{R} q_i\, 2^{R-i}, \qquad q_i \in \{0, 1\}, \tag{B1}$$

with q_1 the most significant bit and q_R the least significant one. A vector $v(q)$ of length 2^R is thereby a tensor $v(q_1, \dots, q_R)$ with R indices of dimension $d = 2$, its *quantics* form; d is the *physical dimension* of every site. A signed index such as a Fourier mode $n \in [-2^{R-1}, 2^{R-1})$ is stored in *two's complement*, as the bits of n mod 2^R:

$$n = -n_1\, 2^{R-1} + \sum_{i=2}^{R} n_i\, 2^{R-i}, \tag{B2}$$

so n_1 is the sign bit, and the nonnegative modes come first, followed by the negative ones, each $n < 0$ at position $2^R + n$. This is the order in which an FFT returns its modes. "Most significant first" means the chain carries $q_1, q_2, \dots, q_R$ in this order; "least significant first" the reverse.

A tensor train (TT, or matrix-product state, MPS) writes a tensor with L indices as a product of matrices, one per index (*site*),

$$\begin{aligned} v(\sigma_1, \dots, \sigma_L) &= M^{[1]}_{\sigma_1} M^{[2]}_{\sigma_2} \cdots M^{[L]}_{\sigma_L}, \\ M^{[i]}_{\sigma_i} &\in \mathbb{C}^{\chi_{i-1} \times \chi_i}, \qquad \chi_0 = \chi_L = 1. \end{aligned} \tag{B3}$$

The χ_i are the *bond dimensions* ("bonds"); the train stores $\sum_i d\, \chi_{i-1}\chi_i$ numbers instead of d^L. A tensor with an additional dense index (the Tucker indices a, b, c below) is a train whose first or last core carries that index, so that χ_0 or χ_L equals its dimension. The smallest bond χ_i that represents the tensor exactly depends only on the tensor and the site order: it is the rank of the *unfolding* matrix whose rows are indexed by $(\sigma_1, \dots, \sigma_i)$ and whose columns by $(\sigma_{i+1}, \dots, \sigma_L)$. A matrix-product operator (MPO) is the same object for a linear map, $W(\sigma'_1 \dots \sigma'_L; \sigma_1 \dots \sigma_L) = W^{[1]}_{\sigma'_1\sigma_1} \cdots W^{[L]}_{\sigma'_L\sigma_L}$, with cores carrying an input and an output bit. Applying an MPO of bond D_W to a train of bond χ gives, before any compression, a train of bond $D_W\chi$.

For a function of two variables the bit strings are concatenated, and the order matters because the bonds are ranks of unfoldings. *Sequential* order puts all bits of one variable before those of the other ($n_1 \dots n_R\, q_1 \dots q_{R_q}$); *interleaved* order alternates them ($q_1 n_1 q_2 n_2 \dots$), possibly with one variable reversed ($q_1 n_R q_2 n_{R-1} \dots$). Bits that interact should sit next to each other. A smooth function of two variables is best sequential: the single matrix coupling them has low rank, and interleaving spreads it over many bonds. A Fourier-type coupling $e^{2\pi i\, nq/2^R}$ is best interleaved with one variable reversed, because the bit q_i couples only to the bits n_j with $i + j > R$ (Appendix J). For this reason the input $\hat{g}_{q,n}$ of the method is interleaved, with n least significant first, and the output $\hat{\chi}_{Q,\ell}$ is sequential (Appendix M).

The cyclic correlation $\frac{1}{N_q}\sum_q A_q B_{q+Q}$ over the momentum grid, $N_q = 2^{R_q}$, is the contraction of $A(q)$ and $B(q')$ with the tensor $\delta_{q',\,(q+Q) \bmod N_q}$, the *adder*, which is an MPO of bond dimension 2 over the bits of q, q' and Q. Binary addition propagates a carry $c \in \{0, 1\}$ from the

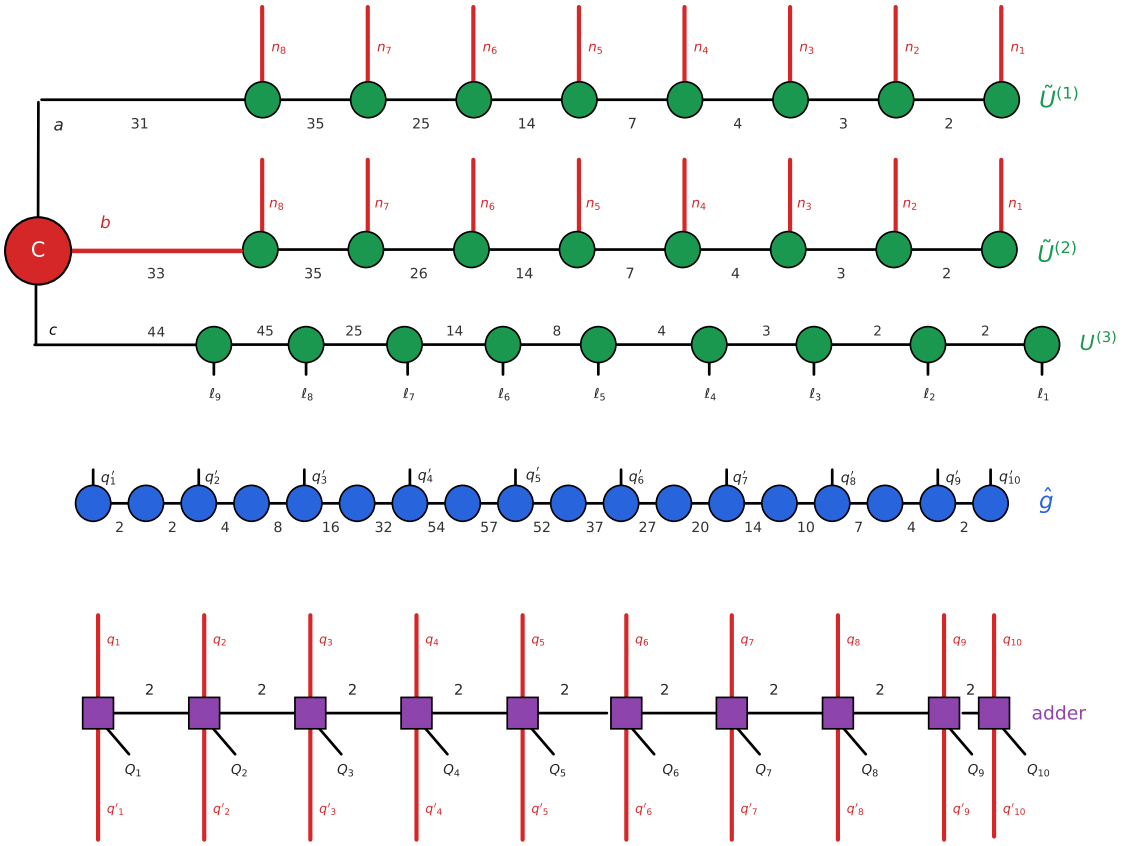


FIG. 7. The ingredients, with the bond dimensions of the run (tolerances in Appendix C). Top: the Tucker tree of the weighted kernel, the core C with the legs a, b, c and the leaves $\tilde{U}^{(1)}$, $\tilde{U}^{(2)}$ over the bits $n_8 \dots n_1$ and $U^{(3)}$ over $\ell_9 \dots \ell_1$. Middle: the input $\hat{g}$, whose sites alternate between momentum bits and bits of n. Bottom: the adder, with the input legs q_i, q'_i and the output legs Q_i; its bond is the carry.

least significant bit upward; the local tensor at bit i is

$$\big[W^{[i]}\big]_{c_{\rm in}c_{\rm out}}(q_i, Q_i, q'_i) = \delta\big(q'_i,\ q_i \oplus Q_i \oplus c_{\rm in}\big) \times \delta\big(c_{\rm out},\ \lfloor (q_i + Q_i + c_{\rm in})/2 \rfloor\big), \tag{B4}$$

with $\oplus$ the sum modulo 2, $c_{\rm in} = 0$ at the least significant bit and the carry out of the most significant bit discarded (addition modulo N_q: momenta are periodic). The bond index is the carry, so the bond is exactly $D_W = 2$ at every site, for any R_q and with no truncation. The *subtractor* $\delta_{q',\,(q-Q) \bmod N_q}$ has the same form with a borrow, $q'_i = q_i \oplus Q_i \oplus b_{\rm in}$ and $b_{\rm out} = 1$ iff $q_i - Q_i - b_{\rm in} < 0$; equivalently, it is the adder with the roles of q and q' exchanged ($q = q' + Q$). In the network (Appendix D) the legs q_i and q'_i are contracted with the two input branches and the legs Q_i remain open, so one contraction gives the correlation for all N_q transfer momenta at once.

The reflection $Q \to -Q \bmod N_q$ (two's-complement negation: complement every bit, then add 1) is also an MPO of bond 2: a bitwise NOT followed by the "+1" tensor $P[c_{\rm out}, {\rm out}, {\rm in}, c_{\rm in}] = 1$ iff ${\rm out} = {\rm in} \oplus c_{\rm in}$ and $c_{\rm out} = {\rm in} \wedge c_{\rm in}$, with $c_{\rm in} = 1$ at the least significant bit and the final carry discarded.

A train produced by an operation (an MPO applied to a train, a sum as in Appendix K, a contraction) in general has bonds larger than needed. A QR sweep followed by a sweep of truncated SVDs, the *recompression*, brings them down to what the tolerance requires, with the error bound of TT-SVD (Appendix L). Every intermediate of the contraction is recompressed this way at relative tolerance 10^{-9}.

To apply an MPO of bond D_W to a train of bond χ, one can form the product train (bond $D_W\chi$) and recompress it, at a cost $O(Ld\,(D_W\chi)^3)$ that is prohibitive when $D_W\chi$ reaches 10^4 (it would be $\approx 13\,800$ in step ④ below). The *zip-up* contracts and truncates site by site in a single sweep. At site i the carried matrix, whose rows are the truncated bond χ'_{i-1}, is contracted with the i-th MPO and train cores, which produces a matrix with $d\chi'_{i-1}$ rows (the truncated bond and the output leg of site i) and $D_W\chi$ columns (the right bonds of the MPO and of the train); a truncated SVD factors it, U becomes the finished core i, and $SV^\dagger$ is carried to site $i+1$. The product of bond $D_W\chi$ is never formed: the largest matrix has one side of size $d\chi'$ and the other $D_W\chi$ (the largest SVD of the run is 550×5832). Since the truncation at site i does not yet know the sites to its right, it is slightly suboptimal; a final recompression sweep removes the ex-

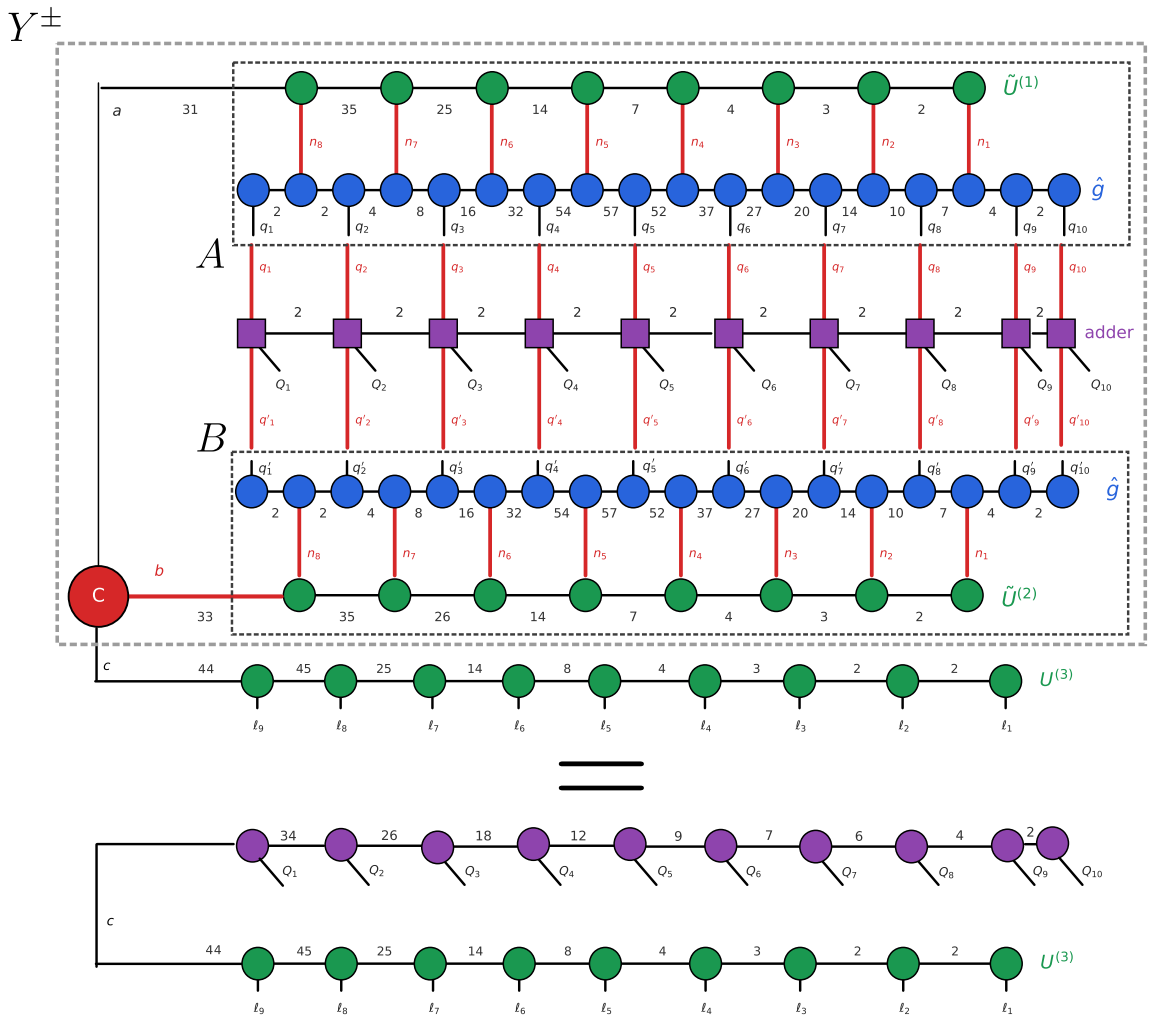


FIG. 8. The network, with the bond dimensions of the run; red legs are contracted. The dashed box A holds $\hat{g}$ and the leaf $\tilde{U}^{(1)}$, contracted over the legs n_i in step ①, and the box B holds $\hat{g}$ and $\tilde{U}^{(2)}$, contracted in step ②. The grey dashed box is the network of step ④, whose result is the train $Y^{\pm}(Q;c)$: A and B joined by the adder over the legs q_i, q'_i, with the core C on their legs a and b. The output leaf $U^{(3)}$ is attached through c. Below the equals sign is the same network after step ④. The drawing shows Y^{+}. For Y^{-} the adder is replaced by the subtractor ($q' = q - Q$, bond: the borrow) and $U^{(3)}$ carries the sign $(-1)^{\ell} = (-1)^{\ell_9}$; the bonds are the same. By (A11), the Y^{-} network equals the Y^{+} network read at $-Q$.

cess (the two rows of bonds in Table I). The same scheme contracts two trains over shared physical legs: in steps ① and ② the leaf train is zipped along $\hat{g}$ over the legs n_i, with an SVD after every q site, and in step ④ the adder sits between A and B', the legs q_i, q'_i are summed and Q_i stays open.

Appendix C: The tensors

The numbers in this and the following sections are for $N_q = 1024$ momenta and $N = 512$ contour points, that is $R_q = 10$ and $R = 9$ bits, with $\lambda = 0.2$ and $\mu = 0.4$. In Fig. 7 and below, all bits are written with the most significant bit first unless stated otherwise; n and ℓ are in two's complement, i.e. FFT order (B2), and bond dimensions are listed from left to right. The input $\hat{g}_{q,n}$ is the raw array (24) as a train; Appendix M explains why the raw array is truncated and the weight $1/s_n$ is put into the leaves. The leaves are the Tucker factors of (A9) written as trains over the bits of n or ℓ, with the Tucker index attached at the least significant bit, where the leaf bonds stay below the Tucker rank. Every tensor is computed to a relative tolerance ε, in the sense of Appendix A 3 for the HOSVD and of Appendix I for the trains:

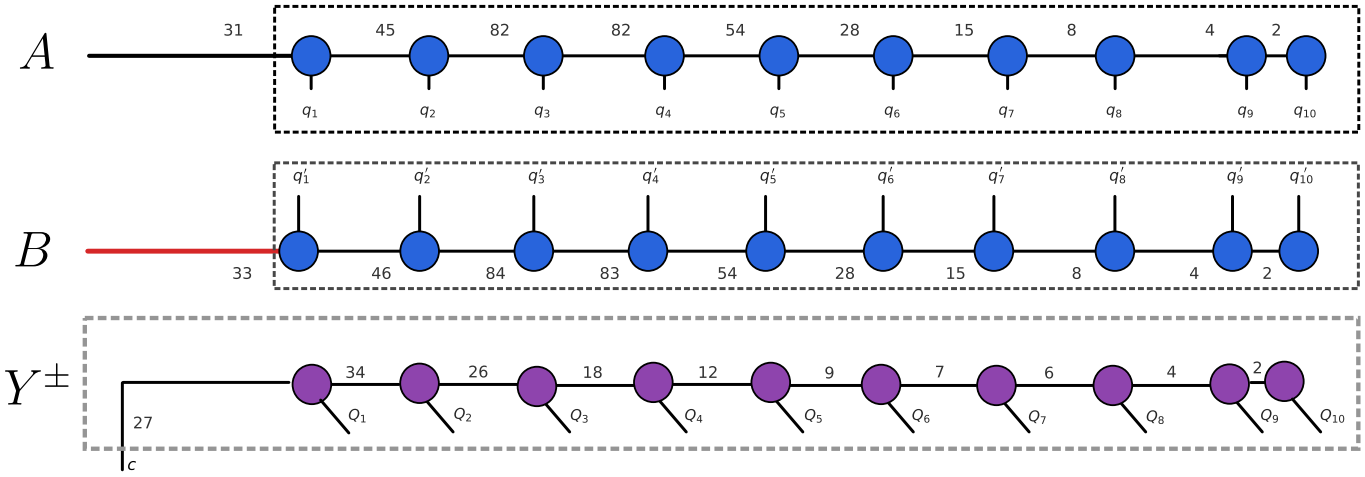


FIG. 9. The intermediate trains after recompression, with the bond dimensions of the run: $A(q;a)$ from step ①, $B(q';b)$ from step ②, and $Y^{\pm}(Q;c)$ from step ④, the same for both signs. Their bonds are the second rows of Table I.

TABLE I. The contraction, step by step. For every step the bonds of the result are given before recompression (first row) and after it (second row). The first, second and fourth steps are zip-ups (Appendix B), so their bonds before recompression are those left by the zip-up. In the cost column χ is the bond dimension of every train and the dimension of the Tucker legs a, b, c; $d = 2$ is the physical dimension of the legs q_i, q'_i, Q_i, n_i, ℓ_i, k_i; $D_W = 2$ is the bond of the adder; $D = 11$ is the bond dimension of the DFT MPO; L is the number of sites.

step	contract	over	result	bonds before recompression → after recompression	cost
①	$\hat{g}$ with $\tilde{U}^{(1)}$	$n_1 \dots n_8$ (zip-up, SVD after every q site)	A: a, $q_1 \dots q_{10}$	31, 50, 100, 193, 293, 139, 47, 10, 4, 2 → 31, 45, 82, 82, 54, 28, 15, 8, 4, 2	$O(Ld^2\chi^4)$
②	$\hat{g}$ with $\tilde{U}^{(2)}$	$n_1 \dots n_8$ (zip-up)	B: b, $q_1 \dots q_{10}$	33, 52, 104, 196, 293, 139, 47, 10, 4, 2 → 33, 46, 84, 83, 54, 28, 15, 8, 4, 2	$O(Ld^2\chi^4)$
③	C with the first site of B	b	B': (a, c), $q_1 \dots q_{10}$	unchanged, no recompression	$O(\chi^4)$
④	A, adder (or subtractor), B'	a, q_i, q'_i (zip-up, SVD after every site)	$Y^{\pm}$: c, $Q_1 \dots Q_{10}$	44, 88, 173, 275, 211, 101, 46, 24, 13, 7 → 27, 34, 26, 18, 12, 9, 7, 6, 4, 2	$O(LD_W d^2\chi^4)$ for each sign
⑤	$U^{(3)}$ with Y^+ and with Y^-, then add	c; sign $(-1)^{\ell}$ on ℓ_9	$\hat{\chi}$: $\ell_1 \dots \ell_9$, $Q_1 \dots Q_{10}$	4, 4, 6, 8, 16, 28, 50, 90, 54, 68, 52, 36, 24, 18, 14, 12, 8, 4 → 2, 2, 2, 3, 5, 9, 16, 24, 24, 34, 26, 18, 12, 9, 7, 6, 4, 2	$O(Ld\chi^3)$
⑥	F with $\hat{\chi}$	$\ell_1 \dots \ell_9$	χ: $k_9 \dots k_1$, $Q_1 \dots Q_{10}$	4, 8, 16, 33, 55, 72, 64, 48, 24, 34, 26, 18, 12, 9, 7, 6, 4, 2 → 2, 4, 7, 8, 11, 16, 20, 24, 24, 34, 26, 18, 12, 9, 7, 6, 4, 2	$O(LdD^3\chi^3)$

TABLE II. Error per transfer momentum $Q = 2\pi m/1024$.

m	error	$\max_k \lvert\chi_{\text{exact}}\rvert$
1	2×10^{-4}	3×10^{-5}
16	8×10^{-7}	6×10^{-3}
64	5×10^{-8}	0.05
160	9×10^{-9}	0.10
256	5×10^{-9}	0.13
512	1.7×10^{-7}	0.22

tensor	computed by	ε
Tucker tree: core C, factors $U^{(i)}$	HOSVD of the weighted kernel	10^{-8}
input $\hat{g}_{q,n}$	TT-SVD of the raw array	10^{-6}
leaves $\tilde{U}^{(1)}$, $\tilde{U}^{(2)}$, $U^{(3)}$	TT-SVD	10^{-10}
inverse DFT F	TT-SVD of the MPO	10^{-12}
adder, subtractor	exact, bond 2	none
intermediates of the contraction	zip-up and recompression	10^{-9}

Appendix D: The contraction

Step ① contracts $\hat{g}$ with the leaf $\tilde{U}^{(1)}$ over the legs $n_1 \dots n_8$, and step ② contracts it with $\tilde{U}^{(2)}$; these are the dashed boxes A and B of Fig. 8, with the contracted

legs in red. The core C is then contracted with the first site of B over b (step ③), which gives B'. The grey dashed box is the network of step ④: A and B' joined by the adder (for Y^+) or the subtractor (for Y^-) over the legs q_i and q'_i, with only c and $Q_1 \dots Q_{10}$ left open. Its result is the train $Y^{\pm}(Q;c)$, drawn below the equals sign, still attached to the output leaf $U^{(3)}$ through c. Step ⑤ does this last contraction for both signs, with the sign $(-1)^{\ell}$ on ℓ_9 for Y^-, and adds the two branches. Step ⑥, the inverse DFT $\ell \to k$ (Appendix J), is not drawn. Figure 9 shows the recompressed trains A, B and $Y^{\pm}$, and Table I gives the bonds and the cost of every step.

With both zip-ups of step ④, 29 s each, the contraction takes about 60 s. The largest bond after recompression is 84, in B; from the core on, every bond is at most 34, the bond of the result itself. Three objects are never formed: the weighted input $\hat{g}/s$, which is incompressible; the correlation $M(Q;a,b)$, whose leg has dimension 1023; and the uncompressed products inside the zip-ups, whose bond would reach $\approx 13\,800$ in step ④.

The branch Y^- does not need its own zip-up. By (A11), $Y^-(Q) = Y^+(-Q \bmod N_q)$: the subtractor version of Fig. 8 is the adder version read at the reflected momentum. The reflection $Q \to -Q$ is the bitwise NOT on every Q bit followed by the exact "+1" MPO of bond 2 (Appendix B), applied to the train $Y^+(Q;c)$. The reflection takes 0.01 s, and the reflected Y^+ agrees with Y^- from the subtractor zip-up to 1.7×10^{-10}. We therefore recommend one zip-up for Y^+ (29 s) followed by the reflection; the whole contraction then takes about 31 s instead of 60 s.

Appendix E: Accuracy

Compared with the Lindhard bubble (F3) at all 1024 transfer momenta and all 512 points of C_μ, the network gives

$$\max_{Q,k} |\chi - \chi_{\text{exact}}| \Big/ \max_{Q,k} |\chi_{\text{exact}}| = 1.7 \times 10^{-7}. \qquad \text{(E1)}$$

Table II gives the error per transfer momentum $Q = 2\pi m/1024$, $\max_k |\chi - \chi_{\text{exact}}| / \max_k |\chi_{\text{exact}}|$. The absolute error is about 10^{-8} at every Q; the relative error grows at small Q only because χ_Q itself vanishes there. The contraction adds no error above 10^{-9}: the MPS result and the same factors contracted densely agree to that level. The error comes from the compressions, mainly that of the kernel at 10^{-8}; by Appendix H, a relative truncation of the weighted kernel is directly an error of $\hat{\chi}$.

Appendix F: The Lindhard function

For the model (11), $h(z) = G_q(z)\, G_{q+Q}(z + i\nu_m) = 1/[(z - \varepsilon_q)(z + i\nu_m - \varepsilon_{q+Q})]$ has only two poles, and

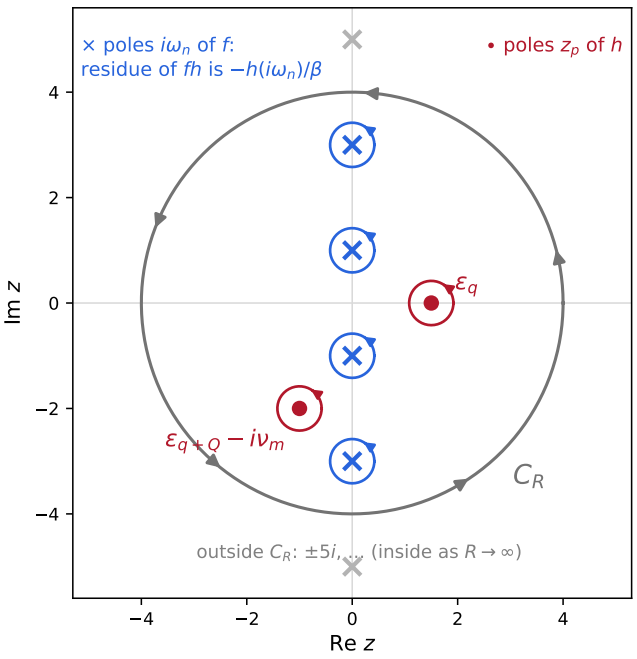


FIG. 10. The residue argument behind (F1), for $\beta = \pi$ and the h of the model with $m = 1$. Crosses: the poles $i\omega_n = \pm i, \pm 3i, \dots$ of f, where fh has the residues $-h(i\omega_n)/\beta$. Dots: the poles of h, ε_q (here 1.5) and $\varepsilon_{q+Q} - i\nu_m$ (here $-1 - 2i$). The integral over the circle C_R equals the sum of the integrals over the small circles, i.e. $2\pi i$ times all residues inside C_R. It tends to zero as $R \to \infty$, so the residues at the crosses and at the dots add up to zero.

the residues at the Matsubara frequencies in (14) can be traded for the residues at these two poles (Fig. 10).

Let h be meromorphic and $O(z^{-2})$ at infinity, with poles z_p none of which is a Matsubara frequency, and let C_R be the circle $|z| = R$, counterclockwise, with R an even multiple of π/β, halfway between two poles of f. By the residue theorem, $\frac{1}{2\pi i} \oint_{C_R} f h\, dz$ is the sum of the residues of fh inside C_R: those at the poles $i\omega_n$ of f and those at the poles z_p of h. On C_R, $|h| \le c/R^2$ and $|f|$ stays bounded ($f \to 1$ for $\operatorname{Re} z \to -\infty$, $f \to 0$ for $\operatorname{Re} z \to +\infty$, and C_R passes between the poles), so the integral is at most of order $2\pi R \cdot c/R^2 \to 0$. As $R \to \infty$ every pole ends up inside, so the two groups of residues add up to zero, and with (14)

$$\frac{1}{\beta} \sum_n h(i\omega_n) = \sum_{\text{poles } z_p \text{ of } h} \operatorname*{Res}_{z=z_p} \big[f(z)\, h(z) \big]. \qquad \text{(F1)}$$

For example, $h(z) = 1/[(z-a)(z-b)]$ has the residues $f(a)/(a-b)$ at a and $f(b)/(b-a)$ at b, so

$$\frac{1}{\beta} \sum_n \frac{1}{(i\omega_n - a)(i\omega_n - b)} = \frac{f(a) - f(b)}{a - b}\,: \qquad \text{(F2)}$$

an infinite sum, whose terms fall like $1/n^2$, replaced by two terms. With $a = 1.5$, $b = -1 - 2i$ and $\beta = \pi$ the

right-hand side is $-0.23163+0.18530\,i$, and the sum over $|n|<10^6$ reaches it to 2×10^{-7}. The bubble of the model is this example with $a=\varepsilon_q$ and $b=\varepsilon_{q+Q}-i\nu_m$. With $f(\varepsilon_{q+Q}-i\nu_m)=f(\varepsilon_{q+Q})$ it is the Lindhard function

$$\chi_Q(i\nu_m)=\frac{1}{N_q}\sum_q\frac{f(\varepsilon_q)-f(\varepsilon_{q+Q})}{i\nu_m+\varepsilon_q-\varepsilon_{q+Q}}. \tag{F3}$$

The right-hand side is a rational function of $i\nu_m$; replacing $i\nu_m$ by any $w\notin\mathbb{R}$ gives the analytic continuation of χ_Q off the real axis, as (16) does for a general G (Sec. II B).

Appendix G: Convergence of the dumbbell quadrature

Let Φ be 8-periodic and analytic in the strip $|\mathrm{Im}\,\phi|<a$, and bounded there by M. Shifting the line of integration in the Fourier coefficients $\hat\Phi_n=\frac{1}{8}\int_{-1}^{7}\Phi\,e^{-i\pi n\phi/4}d\phi$ to $\mathrm{Im}\,\phi=\mp a$ (the vertical sides cancel by periodicity) gives $|\hat\Phi_n|\le Me^{-\pi|n|a/4}$. The N-point trapezoidal sum of $\Phi=\sum_n\hat\Phi_n e^{i\pi n\phi/4}$ is exact for $|n|<N$ and aliases the modes $n=pN$ onto the mean, so

$$\begin{aligned}\Delta\phi\sum_j\Phi(\phi_j)-\int_{-1}^{7}\Phi\,d\phi&=8\sum_{p\neq0}\hat\Phi_{pN},\\ \Big|\,\cdot\,\Big|&\le\frac{16M\,e^{-\pi Na/4}}{1-e^{-\pi Na/4}}.\end{aligned} \tag{G1}$$

For the integrands of (21), $\Phi(\phi)=J(\zeta)F(\omega(\zeta))f(\omega(\zeta))$ with $\zeta=\phi+i\tau\lambda$. A complex shift $\phi\to\phi+i\tau\delta$ moves the level from λ to $\lambda+\delta$, so Φ is analytic in the strip $|\mathrm{Im}\,\phi|<a$ with $a=\tau d$, where d is the distance in level from λ to the nearest singularity: the band (level 0, $d=\lambda$), the poles of f (level 1, $d=1-\lambda$), or a singularity of F. With $\alpha=\pi\tau/4$, the error is

$$\varepsilon_{\mathrm{trap}}\approx e^{-\alpha Nd}. \tag{G2}$$

For the integral over z in (16), $F=G_q(z)[G_{q+Q}(z+w)+G_{q-Q}(z-w)]$, and the singularities of F outside the band are the shifted cuts, at level distance $\approx\lambda'_{\min}$ (near the band the level is locally proportional to the height, so the level of $z_j\pm w_k$ above the band equals the level distance of z_j from the shifted cut), hence $d=\min(\lambda,\lambda'_{\min})$. The estimate $e^{-\alpha N\min(\lambda,\lambda'_{\min})}$ matches the measured errors: for $\mu=0.2$, $\lambda=0.12$ and $N=2^8$ it gives $e^{-28.6}=4\times10^{-13}$, and $\lambda=0.2$, $N=2^8$ gives $e^{-7.8}=4\times10^{-4}$. The two distances are balanced at $\lambda\approx\mu/2$, so the levels are chosen in that ratio.

Appendix H: Decay of the coefficients and the weighted kernel

The function $g(\zeta)=G_q(\omega(\zeta))$ is 8-periodic in $\mathrm{Re}\,\zeta$ and analytic in the open strip $0<\mathrm{Im}\,\zeta<2\tau$, i.e. $0<\lambda<2$: ω maps the strip into $\mathbb{C}\setminus[-E,E]$, where G_q is analytic, and at the two poles of ω (at $\zeta=1+i\tau$ and $5+i\tau$) $G_q(\omega)\to0$ is still analytic. The lines $\lambda=0$ and $\lambda=2$ are the boundary, where ω is on the band. The coefficients at level λ are $\hat g_n(\lambda)=\frac{1}{8}\int_{-1}^{7}g(\phi+i\tau\lambda)e^{-i\pi n\phi/4}d\phi$ (the continuum version of (24)). Shifting the line of integration from height $\tau\lambda_0$ to $\tau\lambda$ inside the strip is allowed by Cauchy's theorem (the vertical sides cancel by periodicity) and gives

$$\begin{aligned}&\hat g_n(\lambda)=e^{-\alpha(\lambda-\lambda_0)n}\,\hat g_n(\lambda_0),\\ &0<\lambda,\lambda_0<2,\qquad\alpha=\frac{\pi\tau}{4}.\end{aligned} \tag{H1}$$

Changing the level is a diagonal operation on the Fourier modes. Because $\hat g_n(\lambda_0)$ stays bounded as $\lambda_0\to0^+$ and as $\lambda_0\to2^-$ (the boundary values of G on the band are bounded or integrable),

$$\begin{aligned}|\hat g_n(\lambda)|&\lesssim e^{-\alpha\lambda n}&&(n>0),\\ |\hat g_n(\lambda)|&\lesssim e^{-\alpha(2-\lambda)|n|}&&(n<0):\end{aligned} \tag{H2}$$

positive modes "see" the band at level 0, negative modes its image at level 2. At $\lambda=0.2$ the rates are 0.196 and 1.77 per mode, so the coefficients reach 10^{-15} of their maximum at $n\approx176$ and $n\approx-20$; smaller coefficients are rounding noise of the double-precision data. The same holds for χ on C_μ with μ in place of λ, which sets the ℓ window.

The worst case among all admissible G is a pole on the band, whose coefficients saturate (H2). Define the *weights*

$$s_n=\begin{cases}e^{-\alpha\lambda n}, & n\ge0,\\ e^{-\alpha(2-\lambda)|n|}, & n<0,\end{cases} \tag{H3}$$

and write the bubble as $\hat\chi_{Q,\ell}=\sum_{n'n}\widetilde K_{n'n\ell}\widetilde P^{\pm}_Q(n',n)$ with the *weighted kernel* $\widetilde K_{n'n\ell}=s_{n'}s_n\widehat K_{n'n\ell}$ and $\widetilde P=P/(s_{n'}s_n)$. For every G whose coefficients are bounded by the worst case, $|\hat g_{q,n}|\le c\,s_n$, the entries of $\widetilde P$ are bounded by c^2, so an error $\delta\widetilde K$ of the weighted kernel gives $|\delta\hat\chi_{Q,\ell}|\le c^2\sum_{n'n}|\delta\widetilde K_{n'n\ell}|$. A relative truncation error ε of $\widetilde K$ is therefore directly an error $\approx\varepsilon$ of $\hat\chi$, uniformly over the admissible G. Compressing the unweighted $\widehat K$ would spend rank on the high modes, which never carry signal, and would give no error control. The input window of Appendix A 2 ends where s_n falls below 10^{-8}. The modes left out cannot change $\hat\chi$ at the tolerance of the kernel: their rows in $\widetilde K$ are at most 3.3×10^{-12} of $\|\widetilde K\|_F$. Keeping them would only add noise. In double precision the HOSVD returns their rows of $U^{(i)}$ as rounding noise of size about $2\times10^{-16}\|\widetilde K\|_F/\sigma^{(i)}_{r_i}\approx3\times10^{-8}$, and the factor $1/s_n$ in the leaves turns it into entries of order 10^7 on the window $[-32,159]$, against at most 1.2×10^3 on $[-10,93]$. On the wider window the error is the same, 1.7×10^{-7}, but the largest bonds of A and B grow from 82 and 84 to 103 and 102, and step ④ takes

45 s instead of 29 s. On the window no floor is needed in (H3).

The weighted kernel does not depend on λ. By (H1), the ratios $a_{q,n} \equiv \hat{g}_{q,n}(\lambda)/s_n(\lambda)$ are the coefficients at level 0 for $n \geq 0$ and at level 2 for $n < 0$: level-free Laurent coefficients of G_q, independent of λ. The bubble is $\hat{\chi}_{Q,\ell} = \sum_{n'n} \widetilde{K}_{n'n\ell} \frac{1}{N_q} \sum_q [a_{q,n'} a_{q+Q,n} + (-1)^\ell a_{q,n'} a_{q-Q,n}]$, and neither $\hat{\chi}$ (a contour integral, independent of the contour) nor the a's depend on λ. Hence $\widetilde{K}$ is the same trilinear form for every forward-safe λ, up to the quadrature error $e^{-\alpha N(\lambda'_{\min}-\lambda)}$. Measured at $\mu = 0.4$, the weighted kernels built at $\lambda = 0.2$ and $\lambda = 0.1$ (both $N = 512$) differ by 1.7×10^{-11} of their maximum. Consequently the ranks of the compressed kernel depend on μ, β and ε only, not on λ or N, and the kernel can be built at the cheapest (λ, N) and applied to data at another λ. A small λ makes the margin $\lambda'_{\min} - \lambda$ large, so N is set by the window alone, with the window cut at the target ε. For $\mu = 0.4$ this means $\lambda = 0.1$, $N = 256$ and 2 s instead of 10 s; for $\mu = 0.2$, $\lambda = 0.05$, $N = 512$ and 29 s plus 19 s of HOSVD instead of 108 s plus 120 s, with identical ranks and errors.

Appendix I: TT-SVD and the truncation criterion

Any tensor can be brought into the form (B3) by successive SVDs: reshape v into the $d \times d^{L-1}$ matrix $v_{(1)}$, compute $v_{(1)} = USV^\dagger$, take $M^{[1]} = U$ (its columns are the first bond), reshape $SV^\dagger$ into a $(d\chi_1) \times d^{L-2}$ matrix, and continue. At each bond the singular values are truncated by the *relative Frobenius criterion*: keep the smallest χ_i with

$$\sum_{s>\chi_i} \sigma_s^2 \leq \frac{\varepsilon^2}{L-1} \|v\|^2, \tag{I1}$$

i.e. each bond discards at most the fraction $\varepsilon^2/(L-1)$ of the squared norm. The truncations at different bonds are orthogonal projections, so the discarded weights add and the relative error in Frobenius norm is at most ε; the result is within a factor $\sqrt{L-1}$ of the best train with the same bonds. All tolerances quoted for trains are relative tolerances ε in this sense. The "bond" or "rank" of an object in the tables is the bond dimension it needs at the stated ε.

Appendix J: The discrete Fourier transform as an MPO

For $k = \sum_i k_i 2^{R-i}$ and $\ell = \sum_j \ell_j 2^{R-j}$,

$$\begin{aligned} e^{2\pi i k\ell/2^R} &= \prod_{i,j} \exp\left(2\pi i\, k_i \ell_j\, 2^{R-i-j}\right) \\ &= \prod_{i+j>R} \exp\left(\frac{2\pi i\, k_i \ell_j}{2^{\,i+j-R}}\right), \end{aligned} \tag{J1}$$

because the factors with $i + j \leq R$ equal 1. The most significant bit k_1 therefore couples only to the least significant bit ℓ_R, k_2 to ℓ_{R-1} and ℓ_R, and so on. If site i of the MPO carries the pair (k_i, ℓ_{R+1-i}), i.e. one index in reversed bit order, then k_i couples to the pairs at sites $i' \leq i$ with the phase $2\pi/2^{\,i-i'+1}$, which decreases geometrically with the distance along the chain and is negligible beyond a distance of about $\log_2(1/\varepsilon)$. The interaction is quasi-local and the MPO has bond dimension $O(\log(1/\varepsilon))$. This is the quantum-Fourier-transform circuit, and the reason the QFT returns the modes in bit-reversed order. With both indices in the same order the strongest coupling, $k_1\ell_R$, joins the two ends of the chain and the bond is maximal. The inverse transform of the method, $F(k,\ell) = e^{i\pi\ell\phi_k/4} = e^{-i\pi\ell/4} e^{2\pi i\ell k/N}$, is a phase times (J1); with ℓ most significant first and k least significant first its MPO has bonds $2, 4, 8, 11, 11, 8, 4, 2$ at tolerance 10^{-12}, so its bond dimension is $D = 11$.

Appendix K: Sum of two trains

The sum $v + v'$ of two trains is the train with cores

$$\begin{aligned} M_\sigma^{[1]} \oplus M_\sigma'^{[1]} &= \begin{pmatrix} M_\sigma^{[1]} & M_\sigma'^{[1]} \end{pmatrix}, \\ M_\sigma^{[L]} \oplus M_\sigma'^{[L]} &= \begin{pmatrix} M_\sigma^{[L]} \\ M_\sigma'^{[L]} \end{pmatrix}, \\ M_\sigma^{[i]} \oplus M_\sigma'^{[i]} &= \begin{pmatrix} M_\sigma^{[i]} & 0 \\ 0 & M_\sigma'^{[i]} \end{pmatrix} \quad (1 < i < L), \end{aligned} \tag{K1}$$

a direct sum: the bonds add, $\chi_i + \chi'_i$, and are brought back down by recompression. A factor that acts on one site, such as the sign $(-1)^\ell = (-1)^{\ell_R}$ (a diagonal tensor on the least significant bit of ℓ), is multiplied into the corresponding core before the sum.

Appendix L: Recompression

A train is recompressed in two sweeps. (i) A QR sweep from left to right: reshape the first core into a $(d\chi_0) \times \chi_1$ matrix, factor it as $M^{[1]} = QR$ with Q having orthonormal columns, keep Q as the new core and multiply R into the core to its right; repeat up to the last site. Now every core but the last is left-orthogonal, and the norm of the train sits in the last core. (ii) A sweep from right to left with truncated SVDs: factor the current core (reshaped to $\chi_{i-1} \times (d\chi_i)$) as $USV^\dagger$, truncate by (I1), keep $V^\dagger$ as the new core and multiply US into the core to its left. Because the rest of the train is orthogonal, the singular values of the local matrix are those of the full unfolding at that bond, so each truncation is optimal for its bond, and with (I1) the total relative error is at most ε, as for TT-SVD. The cost is $O(Ld\chi^3)$ for a train of bond χ.

Appendix M: Design choices

1. Truncate the raw $\hat{g}$ and put the weight $1/s_n$ into the leaves. The raw coefficients decay as in (H2), and truncating them at 10^{-6} still gives a bubble accurate to 10^{-9}: the leaves divided by s_n lie in the range of the unweighted kernel, which is bounded, so errors in the high modes are not amplified. The weighted array $\hat{g}/s$ compresses poorly: at 10^{-6} its bond is 99, close to the 104 modes of the window, against 57 for the raw $\hat{g}$. In it every mode of the window has the same size, and for each momentum $\hat{g}_n/s_n$ is close to a plane wave $e^{-i\pi n\phi(\varepsilon_q)/4}$ whose frequency moves nonlinearly with q.

2. Interleave the bits of the input and keep the output sequential (Appendix B, bit orders). For $\hat{g}$, interleaving q and n with n least significant first gives bond 57 at 10^{-6}; all n bits first gives 75, and all q bits first 145. For the output, frequency bits first and then Q gives 25 to 32, while interleaving gives 36 to 63 for $\hat{\chi}$ and 48 to 83 for χ (at 10^{-6} to 10^{-8}). The contraction delivers the output in the best order by construction.

3. Contract the core into B before the adder. Correlating A and B first creates $M(Q; a, b)$, with a front leg of 1023; its zip-up swells to bonds of 600 to 2600, and capping them costs 3 to 13 % accuracy.

4. Put ℓ on the same bits as k. The inverse DFT is then the square, QFT-type transform (J1), with bond at most 11. Keeping only the 128 window modes on 7 bits gives a rectangular map with bond 32.

5. Compute Y^- by reflecting Y^+ (Appendix D): one zip-up instead of two, 31 s instead of 60 s.

6. Choose $\lambda \lesssim \mu/2$, so that the copies $C_\lambda \pm w_k$ stay outside C_λ ($\lambda'_{\min} > \lambda$, Sec. I A), keep the input modes with $s_n \geq 10^{-8}$ (Appendix H), and set the tolerances to 10^{-8} for the kernel, 10^{-6} for the input, 10^{-10} for the leaves, 10^{-12} for the DFT and 10^{-9} for the recompression. The kernel ranks grow like $\log(1/\varepsilon)$ and like $1/\mu$ (and with β), as in (A13). They do not grow with N or λ, and the kernel may be built at a smaller N than the data's (Appendix H).